

Class 08: Breast Cancer Mini Project

Julianne Lee (PID: A17979401)

Table of contents

Background	1
Data Import	1
Exploratory data analysis	2
Principal Component Analysis (PCA)	3
Hierarchical clustering	9
Combining methods	13
Prediction	14

Background

The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class.

Today we will analyze a dataset from fine needle aspiration (FNA) of a breast mass.

Data Import

The data is made available as a CSV file for download. We can read this in using `read.csv()`:

```
fna.data <- "WisconsinCancer.csv"

wisc.df <- read.csv(fna.data, row.names=1)

head(wisc.df,3)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean		
842302	M	17.99	10.38	122.8	1001		
842517	M	20.57	17.77	132.9	1326		
84300903	M	19.69	21.25	130.0	1203		
		smoothness_mean	compactness_mean	concavity_mean	concave.points_mean		
842302		0.11840	0.27760	0.3001	0.14710		
842517		0.08474	0.07864	0.0869	0.07017		
84300903		0.10960	0.15990	0.1974	0.12790		
		symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se	
842302		0.2419		0.07871	1.0950	0.9053	8.589
842517		0.1812		0.05667	0.5435	0.7339	3.398
84300903		0.2069		0.05999	0.7456	0.7869	4.585
		area_se	smoothness_se	compactness_se	concavity_se	concave.points_se	
842302		153.40	0.006399	0.04904	0.05373	0.01587	
842517		74.08	0.005225	0.01308	0.01860	0.01340	
84300903		94.03	0.006150	0.04006	0.03832	0.02058	
		symmetry_se	fractal_dimension_se	radius_worst	texture_worst		
842302		0.03003		0.006193	25.38	17.33	
842517		0.01389		0.003532	24.99	23.41	
84300903		0.02250		0.004571	23.57	25.53	
		perimeter_worst	area_worst	smoothness_worst	compactness_worst		
842302		184.6	2019	0.1622	0.6656		
842517		158.8	1956	0.1238	0.1866		
84300903		152.5	1709	0.1444	0.4245		
		concavity_worst	concave.points_worst	symmetry_worst			
842302		0.7119		0.2654	0.4601		
842517		0.2416		0.1860	0.2750		
84300903		0.4504		0.2430	0.3613		
		fractal_dimension_worst					
842302		0.11890					
842517		0.08902					
84300903		0.08758					

Make sure we remove or exclude the `diagnosis` column from the dataset that we use for further analysis - this is the expert diagnosis as either M or B:

```
wisc.data <- wisc.df[,-1]
diagnosis <- as.factor(wisc.df$diagnosis)
```

Exploratory data analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.df)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B   M  
357 212
```

```
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

We can use the `grep()` function to help us here:

```
length(grep("_mean", colnames(wisc.df), value=TRUE))
```

```
[1] 10
```

Principal Component Analysis (PCA)

We need to scale our data before PCA with the `scale=TRUE` argument to `prcomp()`.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)  
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010

	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523

Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

0.4427

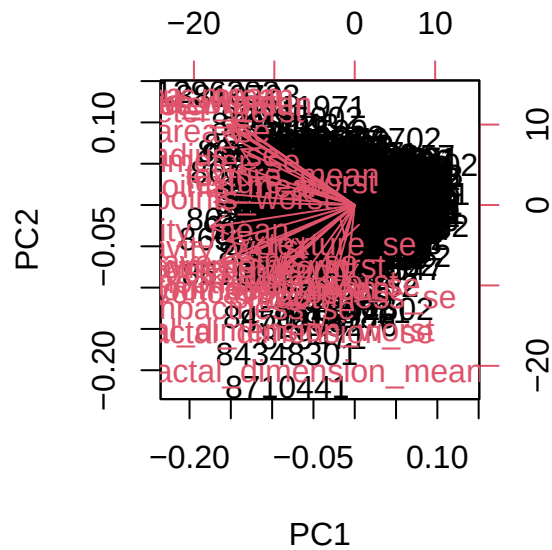
Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7

```
biplot(wisc.pr)
```



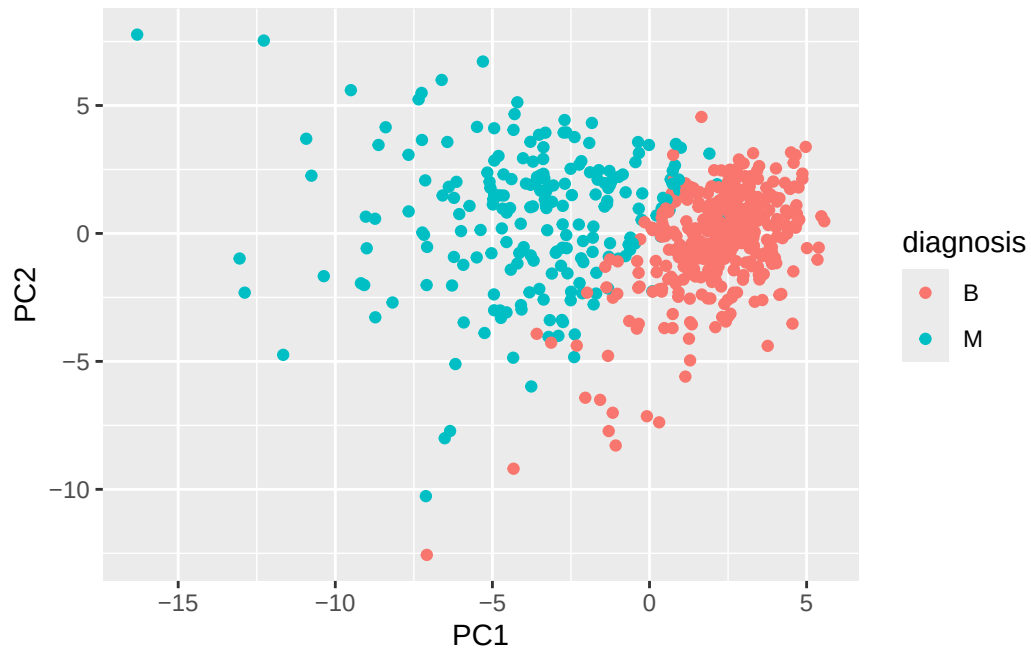
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

Nothing really stands out in this plot, being difficult to understand. All the points and labels are clustered, and the diagnoses are not available.

Let's see the "PC score plot"

```
library(ggplot2)

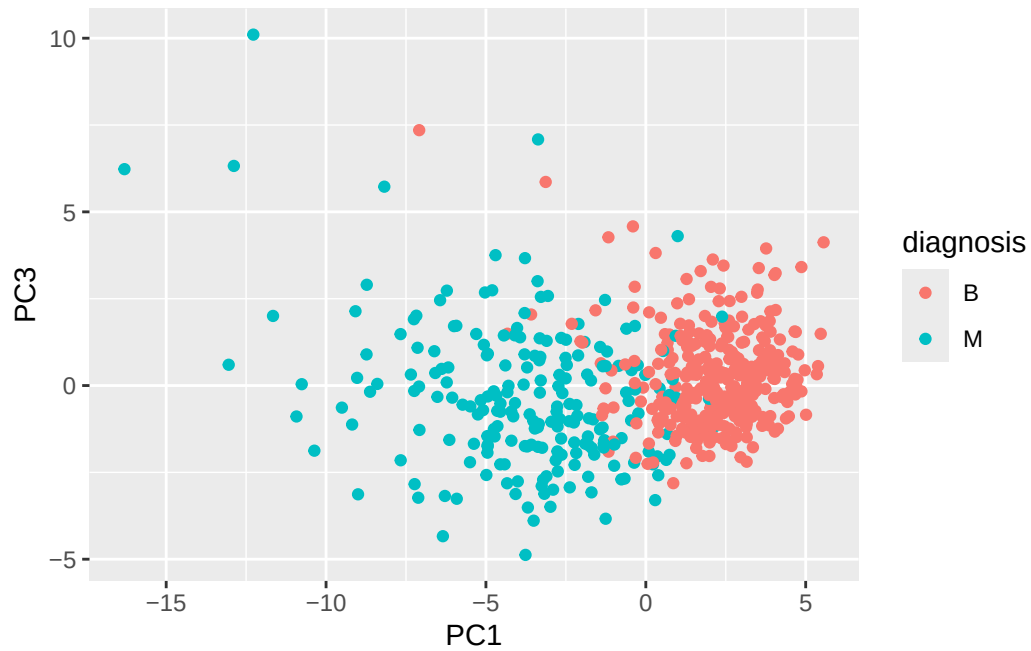
ggplot(wisc.pr$x) +
  aes(PC1, PC2, col=diagnosis) +
  geom_point()
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

PC1 is well capturing a separation of malignant (turquoise) from benign (red) samples.

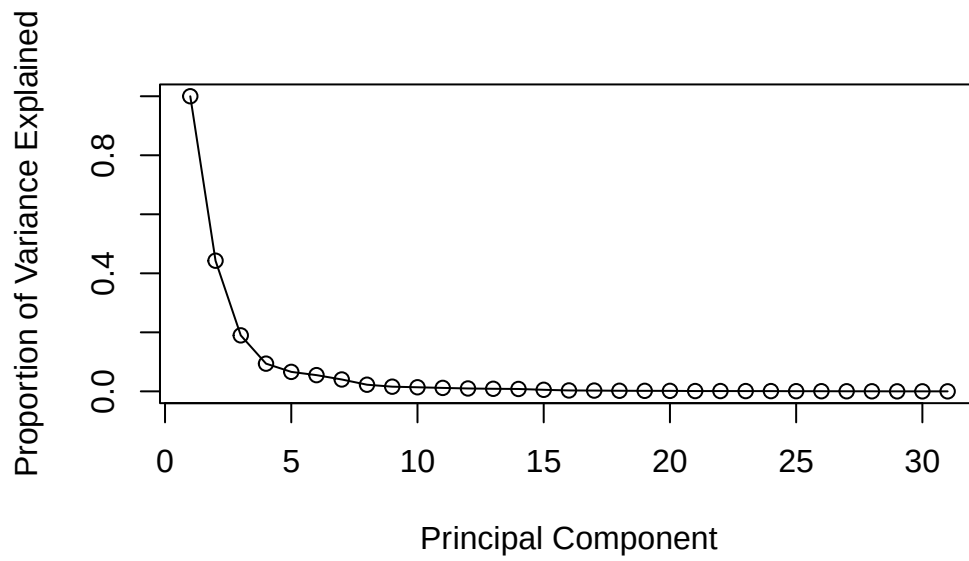
```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



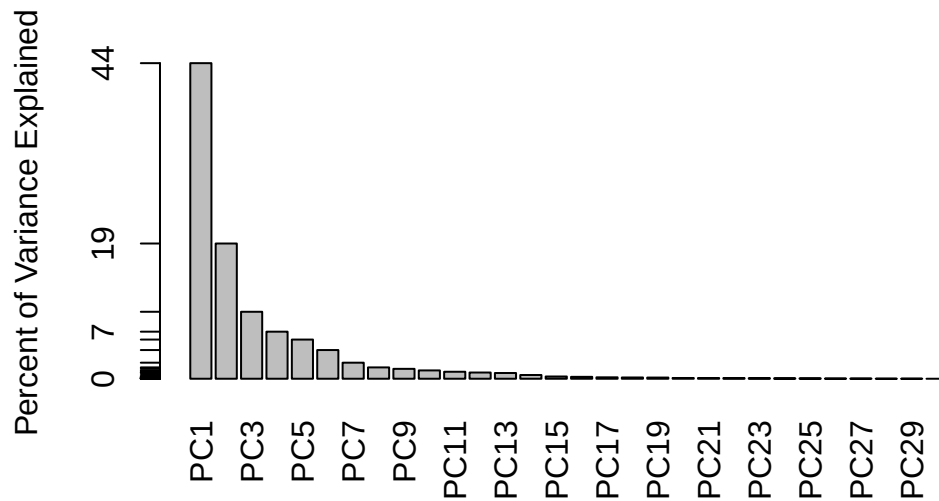
```
pr.var <- wisc.pr$sdev^2  
head(pr.var)
```

```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
pve <- pr.var / sum(pr.var)  
  
plot(c(1,pve), xlab = "Principal Component",  
      ylab = "Proportion of Variance Explained",  
      ylim = c(0, 1), type = "o")
```



```
barplot(pve, ylab = "Percent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

No, `concave.points_mean` has the largest contribution

```
wisc.pr$rotation[,1]["concave.points_mean"]
```

```
concave.points_mean
-0.2608538
```

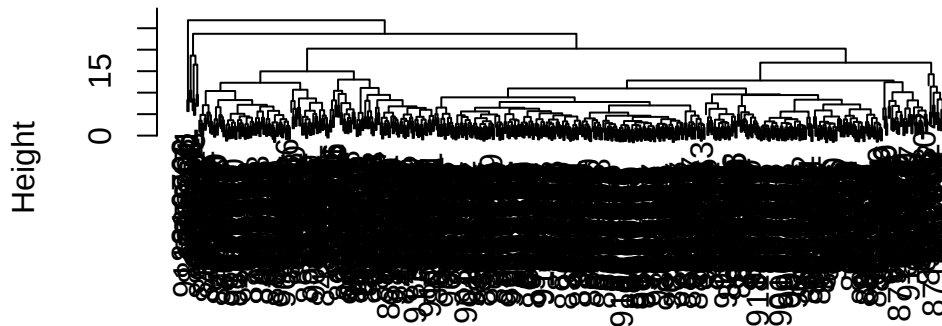
```
sort(abs(wisc.pr$rotation[,1]), decreasing = TRUE)
```

concave.points_mean	concavity_mean	concave.points_worst
0.26085376	0.25840048	0.25088597
compactness_mean	perimeter_worst	concavity_worst
0.23928535	0.23663968	0.22876753
radius_worst	perimeter_mean	area_worst
0.22799663	0.22753729	0.22487053
area_mean	radius_mean	perimeter_se
0.22099499	0.21890244	0.21132592
compactness_worst	radius_se	area_se
0.21009588	0.20597878	0.20286964
concave.points_se	compactness_se	concavity_se
0.18341740	0.17039345	0.15358979
smoothness_mean	symmetry_mean	fractal_dimension_worst
0.14258969	0.13816696	0.13178394
smoothness_worst	symmetry_worst	texture_worst
0.12795256	0.12290456	0.10446933
texture_mean	fractal_dimension_se	fractal_dimension_mean
0.10372458	0.10256832	0.06436335
symmetry_se	texture_se	smoothness_se
0.04249842	0.01742803	0.01453145

Hierarchical clustering

```
wisc.hclust <- hclust(dist(scale(wisc.data)))
plot(wisc.hclust)
```

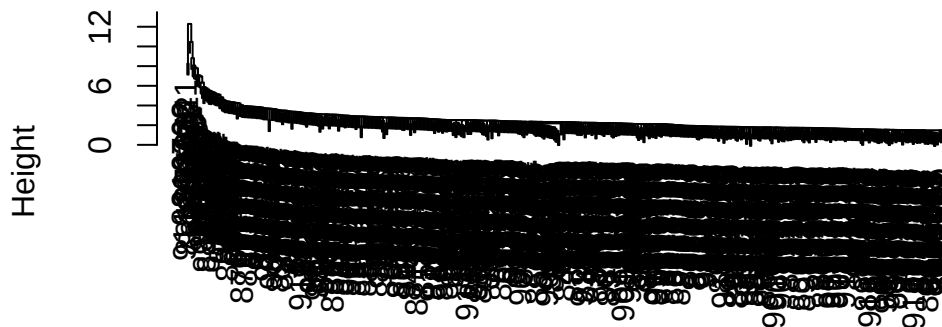
Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust (*, "complete")
```

```
hc.single <- hclust(dist(scale(wisc.data)), method = "single")  
plot(hc.single)
```

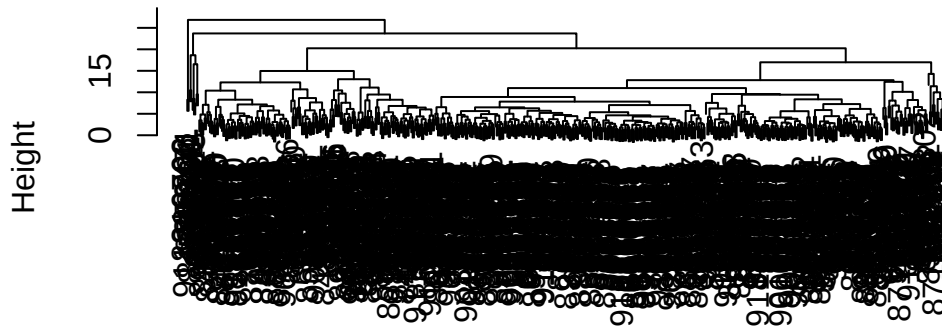
Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust (*, "single")
```

```
hc.complete <- hclust(dist(scale(wisc.data)), method = "complete")  
plot(hc.complete)
```

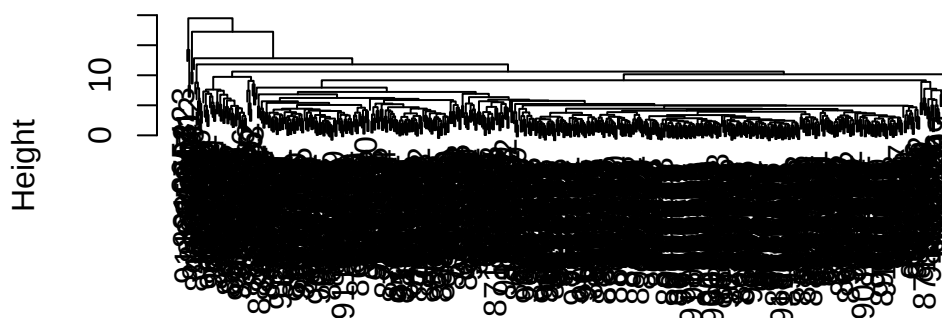
Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust (*, "complete")
```

```
hc.average <- hclust(dist(scale(wisc.data)), method = "average")  
plot(hc.average)
```

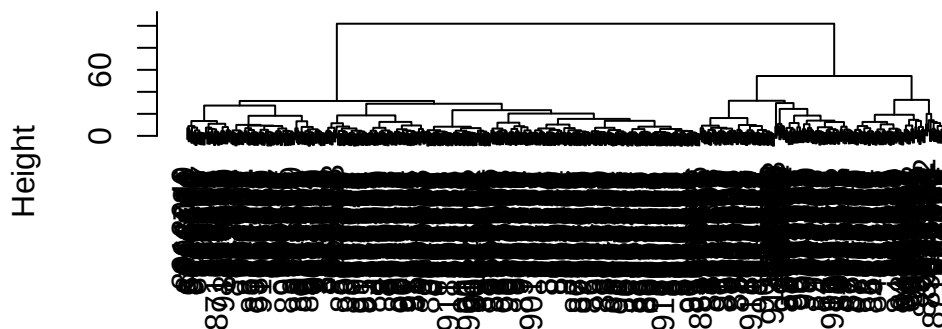
Cluster Dendrogram



```
dist(scale(wisc.data))  
hclust (*, "average")
```

```
hc.ward <- hclust(dist(scale(wisc.data)), method = "ward.D2")  
plot(hc.ward)
```

Cluster Dendrogram



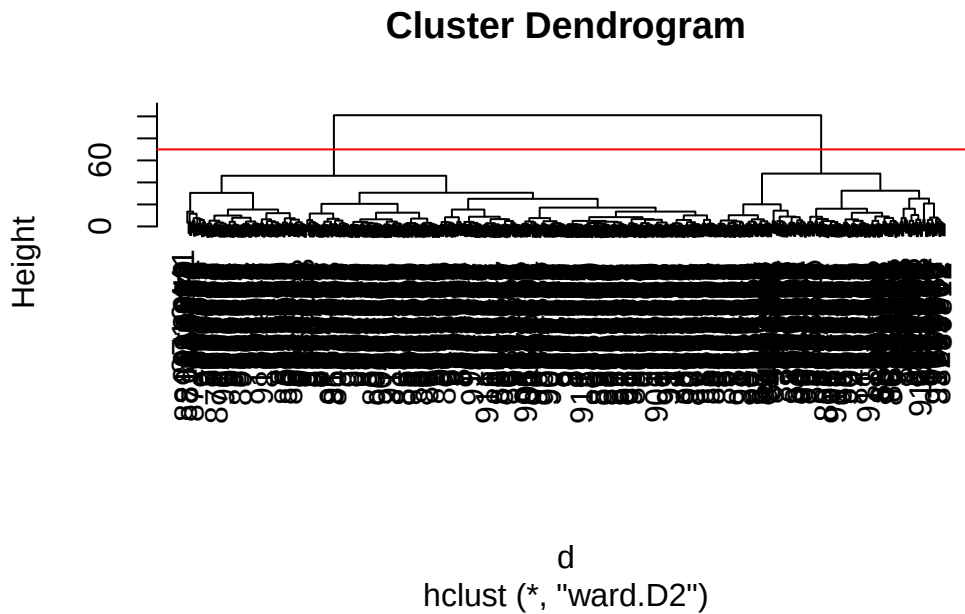
```
dist(scale(wisc.data))  
hclust (*, "ward.D2")
```

Q12. Which method gives your favorite results for the same data.dist dataset? Explain your reasoning.

ward.D2 method gives me favorite result because it gives balanced clusters than the others. This makes it easier to compare the benign and malignant clusters.

Combining methods

```
d <- dist(wisc.pr$x[,1:4])
wisc.pr.hclust <- hclust(d, method="ward.D2")
plot(wisc.pr.hclust)
abline(h=70, col="red")
```



```
grps <- cutree(wisc.pr.hclust, h=70)
table(grps)
```

```
grps
  1   2
171 398
```

How does this clustering `grps` correspond to the expert `diagnosis`?

```
table(diagnosis)
```

```
diagnosis
  B    M
357 212
```

```
table(diagnosis, grps)
```

```
      grps
diagnosis  1    2
  B      6 351
  M     165  47
```

```
wisc.hclust.clusters <- cutree(wisc.hclust, k = 4)
table(wisc.hclust.clusters, diagnosis)
```

```
      diagnosis
wisc.hclust.clusters  B    M
      1  12 165
      2   2   5
      3 343  40
      4   0   2
```

Q14. How well do the hierarchical clustering models you created in the previous sections (i.e. without first doing PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.hclust.clusters` and `wisc.pr.hclust.clusters`) with the vector containing the actual diagnoses.

While both performed well, PCA-based clustering was slightly better than the original hierarchical clustering at separating the benign and malignant diagnoses.

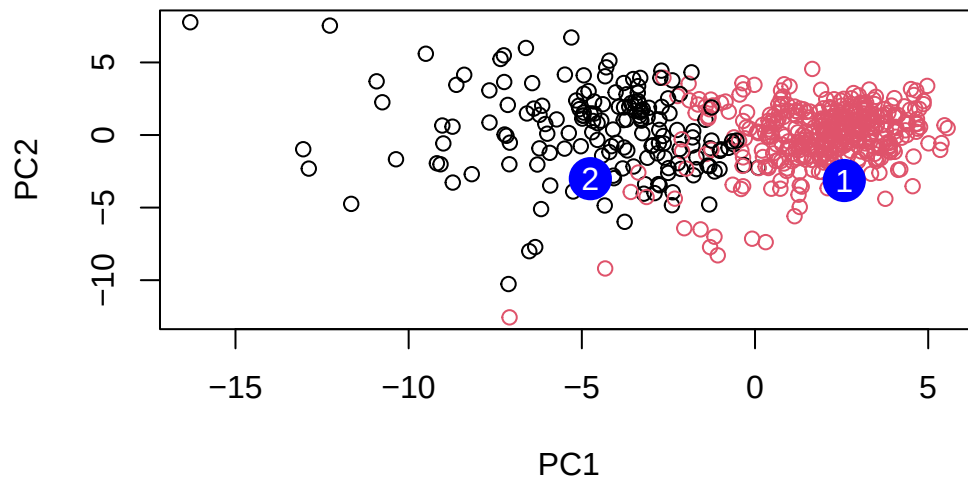
Prediction

We will use the `predict()` function that will take our PCA model from before and new cancer cell data and project that data onto our PCA space.

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results?

Patient 2 as it clusters more with malignant group.